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Week 4 Project Summary

AnalystLab Africa — Experience Lab | HealthConnect Clinic Project | Data Analytics Track

1. Problem Addressed

HealthConnect Clinic is experiencing a high rate of missed appointments (no-shows). As the Data Analytics track, my role this week was to understand the appointment dataset and identify the patterns and factors associated with no-shows, so the clinic can eventually make data-informed decisions to reduce missed appointments and improve appointment slot efficiency.

2. Resources Used

- HealthConnect Appointment Dataset (HealthConnect_Appointment_Data.csv) — 5,000 fictional appointment records across 18 columns.
- HealthConnect Data Dictionary — used to understand the meaning of each column before analysis.

3. Key Observations from Week 4

- The dataset contains two columns with missing values — Distance_to_Clinic_KM and Waiting_Time_Minutes — while the rest of the dataset appears structurally consistent, with no duplicate rows identified.
- The dataset includes a Patient ID field that allows individual patients to be tracked across multiple appointments, which supports deeper analysis such as linking a patient's waiting time to their attendance at a future visit.
- Five relevant business questions were identified, covering distance, reminders, waiting time, previous no-show history, and demographics, each of which can realistically be answered using the available columns.

4. Proposed Approach

The proposed approach involves grouping patients by key factors — distance range, reminder status, prior waiting time, and previous no-show history — and comparing no-show rates across these groups to identify which factors show the strongest association with missed appointments. For questions requiring a patient's appointment history to be linked across visits, derived columns (such as appointment sequence number and next-appointment outcome) will be created from existing fields.

5. Key Considerations

- Missing values in two columns will need to be addressed before those specific factors can be reliably analyzed.
- The dataset is fictional; findings may not fully reflect real-world patient behavior.
- Some business questions (e.g., linking waiting time to a patient's next appointment) require additional data preparation before analysis can proceed.

6. Proposed Focus for Week 5

In Week 5, the focus will shift toward cleaning the identified data quality issues, calculating the proposed KPIs, and beginning the comparative analysis outlined in the proposed approach, starting with the most actionable questions — the relationship between reminders and no-shows, and between distance and no-shows.